

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	SDL 2023
People present	APA, BAY, JJ, ARJ, Reinhard Drews + Gunnar Björn
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	15 JUN

Notes: Upgrade from titanium to 100 mm alu mast
Check loggerprogram should be "GC_NETv2_4.cr1x"

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	No			Clean? Y / N
Iridium modem IMEI number	Y (inside new loggerbox)		S: 300534063815680		
Inclinometer/compass	N	No	Attached to the radiometer? Y / N		
Satellite antenna	N	No			
Wind sensor upper (Or maybe for lower)	Y	No	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Wind sensor lower	N	No	Black box pointing same direction as the radiometer? Y / N		
Temperature / humidity assembly upper	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Temperature / humidity assembly lower	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper	Y	Y	Old sensor with new membrane? Y / N		
Sonic ranger lower	Y	Y	Old sensor with new membrane? Y / N		
Thermistor string	Y	Y			
Solar panel	Y	Y	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	Y	Y	At top of logger enclosure? Y / N		
Logger box ID	Y (M3-0006)	Y	CR1000 or CR1000X? ID number:		
Logger enclosure	Y	Y	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y	Y	New box? Y / N New desiccant bags? Y / N		
Mast extension inserted?	N	N	How many inserted? How long? Notes:		
Mast (Complete set)	Y	Y	New mast? Y / N New sensor boom? Y / N		

Station name	SDL 2023
Date	15 JUN

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	—	None
Download data to PC and/or change SD card (wait for green light)	— Y / N	Y / N
Name of logger program	No Program	GC-VET 2.4
Screen dump of values in fast scan mode	— Y / N	Y / N
Photos of mast, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if looking up)	4 °	-2 °
Mast tilt across boom looking from radiometer (- if clockwise)	1 °	0 °
Upper boom direction with radiometer relative to north	° true/magn.	0 ° true/magn.
Lower boom perpendicular to upper boom. If not, measure.	Y / N	Y / N
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	Y / N	Y / N
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	Y / N	Y / N
Upper temperature / humidity height (bottom of casing – surface)	130 cm	425 123 cm
Lower temperature / humidity height (bottom of casing – surface)	77 cm	250 cm
Upper sonic ranger height (bottom of casing – surface)	117 cm	377 cm
Lower sonic ranger height (bottom of casing – surface)	37 cm	256 cm
Length of thermistor string on surface from surface marking	— cm	0 cm
Apparent damage to sensors? If yes, which?	Y / (N)	Y / N

Notes:

Power cable loosened at plug in logger box
1st sensor at surface

On arrival (old titanium station)
- Battery state NAN

Must-do at next visit:

- L

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	SDM 2023
People present	APA RAV, ARV, JJ, Perla & Drews, Ege & Neso
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	11 JUN 2023

Notes: Upgrade from titanium to 100 mm alu mast

Check loggerprogram in shipped loggerbox, should be "GC_NETv2_4.cr1x"

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	N			Clean? ☒ Y / N
Iridium modem IMEI number	Y	Y	S: 300234065154530		
Inclinometer/compass	N	N	Attached to the radiometer? ☒ Y / N		
Satellite antenna	N	N			
Wind sensor upper (Or maybe for lower)	Y	Y	Black box pointing toward the radiometer (i.e., along the boom)? ☒ Y / N		
Wind sensor lower	N	N	Black box pointing same direction as the radiometer? ☒ Y / N		
Temperature / humidity assembly upper	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Temperature / humidity assembly lower	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper Incl cable for one of them	Y	Y cable reused	Old sensor with new membrane? Y / N		
Sonic ranger lower	Y	Y cable reused	Old sensor with new membrane? Y / N		
Thermistor string	Y	Y			
Solar panel	Y	Y	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	Y	Y	At top of logger enclosure? Y / N		
Logger box ID	M3-0002	Y	CR1000 or CR1000X? ID number:		
Logger enclosure	Y	Y	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y	Y	New box? Y / N New desiccant bags? Y / N		
Mast extension inserted?	N	N new mast	How many inserted? How long? Notes:		
Mast (Complete set)	Y	Y	New mast? Y / N New sensor boom? Y / N		

Station name	SDM 2023
Date	11 JUN APA, BAV, ARV, TO, Reinhard Druss

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data to PC and/or change SD card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of mast, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if looking up)	(it is looking down) + 4 °	0 °
Mast tilt across boom looking from radiometer (- if clockwise)	+ 3 °	0 °
Upper boom direction with radiometer relative to north	° true/magn.	0 ° true/magn.
Lower boom perpendicular to upper boom. If not, measure.	(Y) / N °	(Y) / N °
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	(Y) / N °	(Y) / N °
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	Y / N °	(Y) / N °
Upper temperature / humidity height (bottom of casing – surface)	70 cm cm	431 cm
Lower temperature / humidity height (bottom of casing – surface)	cm	295 cm
Upper sonic ranger height (bottom of casing – surface)	56 cm	395 cm
Lower sonic ranger height (bottom of casing – surface)	cm	253 cm
Length of thermistor string on surface from surface marking	cm	top sensor at surface cm
Apparent damage to sensors? If yes, which?	(Y) / N	Y / (N)

Notes:

Sandbags ~ 10 cm, mast drift sandbags 15-20 cm

→ to lowest rib in casing

- all three tether wires intact (tight)
- Solar panel 1/3 covered (drift)

lower Young broken

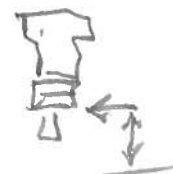
Must-do at next visit:

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	NSE 2023
People present	APA, JJ, BAV, ARD, Reinhard Drows
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	9 JUN

Notes: Upgrade from titanium to 100 mm alu mast
Check loggerprogram should be "GC_NETv2_4.cr1x"

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N				Clean? Y / N
Iridium modem IMEI number	Y (inside new loggerbox)		S: 300234065154530		
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	N				
Wind sensor upper (Or maybe for lower)	Y		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Wind sensor lower	N		Black box pointing same direction as the radiometer? Y / N		
Temperature / humidity assembly upper	Y		Fan spins? Y / N Sounds, OK? Y / N		
Temperature / humidity assembly lower	Y		Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper Incl cable for one of them	Y		Old sensor with new membrane? Y / N		
Sonic ranger lower	Y		Old sensor with new membrane? Y / N		
Thermistor string	Y				
Solar panel	Y		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	Y		At top of logger enclosure? Y / N		
Logger box ID	Y (M3- 0003)		CR1000 or CR1000X? ID number:		
Logger enclosure	Y		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y		New box? Y / N New desiccant bags? Y / N		
Mast extension inserted?	N		How many inserted? How long? Notes:		
Mast (Complete set)	Y		New mast? Y / N New sensor boom? Y / N		



Station name	NSE 2023	Eggert & Neso
Date	9 JUN ~ APR BAY JJ BRU R. DEUS	

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	DEA	0
Download data to PC and/or change SD card (wait for green light)	Y/N	Y/N
Name of logger program		see other por 2.4
Screen dump of values in fast scan mode	DEAD Y/N	Y/N
Photos of mast, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if looking up)	-2°	°
Mast tilt across boom looking from radiometer (- if clockwise)	-2°	°
Upper boom direction with radiometer relative to north	° true/magn.	0° true/magn.
Lower boom perpendicular to upper boom. If not, measure.	clockw. 5° from above Y/N	Y/N
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	Y/N	Y/N
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	Y/N	Y/N
Upper temperature / humidity height (bottom of casing – surface)	70 cm	356 cm
Lower temperature / humidity height (bottom of casing – surface)	0 cm	299 cm
Upper sonic ranger height (bottom of casing – surface)	35 cm	356 cm
Lower sonic ranger height (bottom of casing – surface)	(2.15) cm	240 cm
Length of thermistor string on surface from surface marking	N/A cm	cm
Apparent damage to sensors? If yes, which?	see notes Y/N	Y/N

Notes:

ARR 299
Upper wind sensor 79 cm above surface (to axis) / 299
Lower wind sensor 25 cm
Radiometer 70 cm

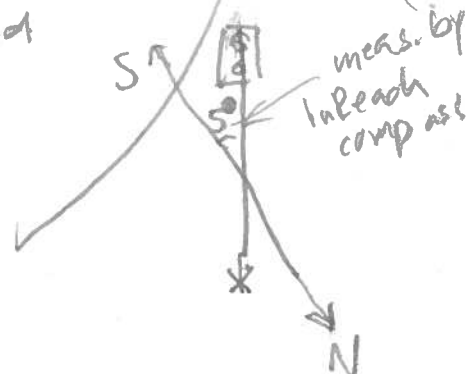
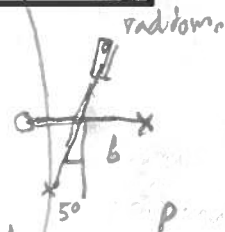
Sustrugi ~10 cm

Radiometer upper SW dome slightly misty
Snow filled up lower Luftt

Must do at next visit:

Radiometer turned a bit according to mast
tilt is +3° 3° seen from end
pointing upwards 10

First sensor
at surface



Maintenance sheet version 10 for ablation area (tripod, one level)

Station name	KAN_U 2023
People present	APA, JJ, BAV, ABU, Reinhard Drews (Eggert + Neso)
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	8 JUNE 2023 9:30 LT

Notes: Make mast solution, but promise v2 instrumentation and GC net batterybox

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	67 <i>Y</i>	202705 <i>calib Nov 2020</i>	213270	Clean? <i>Y</i> /N <i>Calib date March 2023</i>
Ablation hose assembly	N				No air in hose and bladder half full? Y/N
Iridium modem IMEI number (inside loggerbox)	Y		S: 30053406245410		<i>external still in use</i>
Inclinometer/compass	N	<i>Y+N</i>	Attached to the radiometer? Y / N <i>#1 old inclin. still in use</i> <i>#2: Inside, but not used due to v.2</i>		
Satellite antenna	N				
Wind sensor	Y	<i>104 axis cm</i>	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly	Y		Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Lufft sensor (Y / N)		
Sonic ranger on AWS	Y	<i>56 cm</i>	Old sensor with new membrane? Y / N		
Sonic ranger on stake(s)	Y	<i>0 cm</i>	Old sensor with new membrane? Y / N NB: make sure not being buried next winter		
Thermistor string	Y				
Solar panel	Y		Clean? Y / N NB: is not effective when partially covered		
GPS antenna (inside loggerbox)	Y		At top of logger enclosure? Y / N		
Logger box ID	T2-0002		CR1000 or CR1000X? ID number:		
Multiplexer	N		Clicking during measurement cycle? Y / N		
Logger enclosure (v2)	Y		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	Y		New box? Y / N New desiccant bags? Y / N		
Stakes	Y		New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
New complete mast set, incl 1xboom	Y		New tripod? Y / N New sensor boom? Y / N		

Station name	KAN_U 2023
Date	8 JUNE 2023

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	-03:00	
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program	Promice2015e.Cr1	Promice 2022 v1.1
Screen dump of values in fast scan mode	☒ Y / N	☒ Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	0 °	+ 0.5 °
Mast tilt across boom looking from radiometer (- if clockwise)	0 °	0 °
Boom direction relative to north	° true/magn.	° true/magn.
Box on wind sensor exactly towards mast? If not, measure.	☒ Y / N	☒ Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	tilt - 1° Y / N radiom 0°	☒ Y / N
Temperature / humidity height (bottom of casing – surface)	54 cm	330 cm
Stake assembly standing?	Y / N / (sort of)	Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	45° tilt 115 cm	cm
Free length of stake 2 (if 3: middle)	45° tilt 125 cm	cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	cm
Length of thermistor string on surface from surface marking	0 cm from 1st thermistor	cm
Apparent damage to sensors? If yes, which?	☒ Y / N	Y / N

Notes:

sardugi : 3-4 cm

solar panel partly buried

GPS mount bottom to mast connector 9cm

Young wind sensor broken (head off)

NB : We forgot to refill the thermistor string hole with snow

— hangs in hole, but closed at to top — most thermistor placed at surface

Must do at next visit:

lower SPSA 240 cm

upper SPSA 140 + (280 - 60)

Ratronic bottom of casing : 330 cm

Radiometer : 357 cm

Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	DY2 2023
People present	APD, NBK, ZAV, AEU, JJ, Reinhard Drews, Gummert & Bjorn
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	19 JUN

Notes: Raise mast of needed

calib. date
March 20

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	Y		131455	Clean? Y / N
Iridium modem IMEI number	N	N		300534063811780	
Inclinometer/compass	N	Y	Attached to the radiometer? Y / N		
Satellite antenna	N	N			
Wind sensor upper	N	N	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Wind sensor lower	N	N	Black box pointing same direction as the radiometer? Y / N		
Temperature / humidity assembly upper	Y	Y	Fan spins? Y / N Sounds, OK? Y / N cannot hear		
Temperature / humidity assembly lower	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper	Y	Y	Old sensor with new membrane? Y / N		
Sonic ranger lower	Y	Y	Old sensor with new membrane? Y / N		
Thermistor string	N	N			
Solar panel	N	N	Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N	N	At top of logger enclosure? Y / N		
Logger box ID	N	N	CR1000 or CR1000X? ID number: No		
Logger enclosure	N	N	Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N	N	New box? Y / N New desiccant bags? Y / N		
Mast extension inserted?	Y	Y	How many inserted? How long? 150 cm Notes:		
Mast	N	N	New mast? Y / N New sensor boom? Y / N		

Station name	DY2 2023
Date	14 JUN APA, NBE, ALY BAY, JJ, Reinhard Drums

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)	-00:46	-00:46 (same)
Download data to PC and/or change SD card (wait for green light)	Y/N	Y/N
Name of logger program	GC-Net v2	GC-Net v2.4
Screen dump of values in fast scan mode	Y/N	Y/N
Photos of mast, sensors, logger box wiring etc. (more is better)	Y/N	Y/N
Mast tilt towards radiometer (- if looking up)	-2°	same °
Mast tilt across boom looking from radiometer (- if clockwise)	-1°	same °
Upper boom direction with radiometer relative to north	0° (true/magn.)	0° (true/magn.)
Lower boom perpendicular to upper boom. If not, measure.	Y/N 0°	Y/N 0°
Box on upper wind sensor exactly same direction as radiometer? If not, measure.	Y/N 0°	Y/N 0°
Box on lower wind sensor exactly same direction as radiometer? If not, measure.	Y/N 0°	Y/N 0°
Upper temperature / humidity height (bottom of casing – surface)	324 cm	484 cm
Lower temperature / humidity height (bottom of casing – surface)	207 cm	371 cm
Upper sonic ranger height (bottom of casing – surface)	284 cm	446 cm
Lower sonic ranger height (bottom of casing – surface)	167 cm	324 cm
Length of thermistor string on surface from surface marking	N/A cm	N/A cm
Apparent damage to sensors? If yes, which?	Y/N	Y/N

Notes:

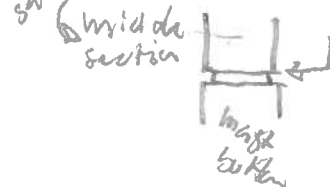
logger box OS: 0502

→ inclinometer not working

Mast top to middle of upper boom (axis center): 11.0 cm

AWSS: Rover 16.7 cm bottom of 90° mount to bottom of ^{old} logger box
 same → section of mast

Must-do at next visit:



Maintenance sheet version 10 for accumulation area (single mast, two levels)

Station name	DY2 old_style_mast 2023
People present	
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	

Notes: Retrieve old equipment and mast (basically everything from OLD station)

AWS to do list		Check after maintenance			
Part	Shipped? (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N				Clean? Y / N
Iridium modem IMEI number	N				
Inclinometer/compass	N		Attached to the radiometer? Y / N		
Satellite antenna	N				
Wind sensor upper	N		Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Wind sensor lower	N		Black box pointing same direction as the radiometer? Y / N		
Temperature / humidity assembly upper	N		Fan spins? Y / N Sounds, OK? Y / N		
Temperature / humidity assembly lower	N		Fan spins? Y / N Sounds, OK? Y / N		
Sonic ranger upper	N		Old sensor with new membrane? Y / N		
Sonic ranger lower	N		Old sensor with new membrane? Y / N		
Thermistor string	N				
Solar panel	N		Clean? Y / N NB: is not effective when partially covered		
GPS antenna	N		At top of logger enclosure? Y / N		
Logger box ID	N		CR1000 or CR1000X? ID number:		
Logger enclosure	N		Replaced (including everything inside)? Y / N New desiccant bags? Y / N		
Battery box	N		New box? Y / N New desiccant bags? Y / N		
Mast extension inserted?	N		How many inserted? How long? Notes:		
Mast	N		New mast? Y / N New sensor boom? Y / N		

Station name	DY2 old_style_mast 2023
Date	14 JUN

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data to PC and/or change SD card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	Y / N
Photos of mast, sensors, logger box wiring etc. (more is better)	☒ Y / N	Y / N
Mast tilt towards radiometer (- if looking up)	- 0 °	°
Mast tilt across boom looking from radiometer (- if clockwise)	°	°
Upper boom direction with radiometer relative to north	2.5 ° true/magn.	° true/magn.
Lower boom perpendicular to upper boom. If not, measure.	Y / N °	Y / N °
Box on upper wind sensor <u>exactly same direction as radiometer?</u> If not, measure. <i>↳ do South ✓</i>	Y / N 0 °	Y / N °
Box on lower wind sensor <u>exactly same direction as radiometer?</u> If not, measure. <i>↳ towards South ✓</i>	Y / N 0 °	Y / N °
Upper temperature / humidity height (bottom of casing – surface)	253 cm	cm
Lower temperature / humidity height (bottom of casing – surface)	139 cm	cm
Upper sonic ranger height (bottom of casing – surface)	235 cm	cm
Lower sonic ranger height (bottom of casing – surface)	181 cm	cm
Length of thermistor string on surface from surface marking	N/A cm	cm
Apparent damage to sensors? If yes, which?	☒ Y / N	Y / N

Notes:

Radiometer angle to



Must-do at next visit:

- ↳ Broken Net Radiometer
- ↳ Upper SRSO membrane bad
- ↳ Lower SRSO membrane bad
- ↳ a bit of ice in bottom of old CIRS station casing